University of Dhaka
Institute of Information Technology
Bachelor of Science in Software Engineering (BSSE)

1st Semester MID Examination, 2024
Course Code: 106 Introduction to Software Engineering
Marks: 20 Time: 1 Hour

Professionalism Excellence Respect

Answer all the questions. The weight of each question is mentioned at the right side

1. Define software. What are the attributes of a good software. 5
2. How you differ software engineering from computer science? What are key challenges facing software engineering. 5
3. Describe the work of three generic faces which are associated with software engineering. 5
4. Define software process? Describe the phases of a problem solving loop with diagram. 5

Good Luck

INSTITUTE OF INFORMATION TECHNOLOGY

Midterm Examination, 2024: BS in Software Engineering

Course title: *Probability and Statistics for Engineers - I* (Stat1103)

Total marks: *20* Time allowed: *60 minutes*

Attempts All the Questions

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1. *(3 points)* Show the following:

   (a) $\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - (\sum_{j=1}^{n} x_j)^2/n.$

   (b) $\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - (\sum_i x_i)(\sum_i y_i)/n.$

   (c) $\sum_{i=1}^{n}(z_i - \bar{z})^2 = (1/b^2)\sum_{i=1}^{n}(x_i - \bar{x})^2$, where $z_i = (x_i - a)/b$

2. *(7 points)* From a sample of midterm exams scores $\{x_1, x_2, \ldots, x_n\}$, we have the following information on raw and summary data.

   raw data, x : 11.5, 11.6, 12.4, 13.2, 13.3, 14.4, 14.9, 15.5, 15.7, 16.5, 17.8, 18.3, 18.8, 19, 19.1
   19.9, 19.9, 19.9, 20.5, 20.5, 20.9, 22.8, 23, 25.1, 27.6

   Summary : $\sum x^2 = 8596.99$, $\sum x = 452.1$, $n = 25$

   (a) Calculate sample mean ($\bar{x}$) and variance (s_x^2) of midterm exam scores. (b) Calculate the sample mean and variance of transformed variable $(x - \bar{x})/s_x$. (c) Calculate three quartiles. (d) Draw a boxplot of the midterm exam score. (e) Make a statement about the shape of the distribution of the midterm exam scores.

3. (a) *(2 points)* Define experiment, sample space, and event with appropriate examples.

   (b) *(2 points)* What are the axioms of probability?

4. *(6 points)* Five independent flips of a coin are made, and $P(H) = 0.6$. Find the probability that

   (a) the first three flips are the same;

   (b) either the first three flips are the same, or the last three flips are the same;

   (c) there are at least two heads among the first three flips and at least two tails among the last three flips.

Student's name and class roll should be written at the top of this question paper and leave it inside the exam script before submission.

- **Name:** Md. Tawnat Hossen Nibir
- **Class roll number:** 25

Section B: 20 (4*5)

1. Write a program where you store all the answers (a, b, c, d or e) given by you from the above Section A. In a separate array the correct answer keys are given. Then
   a. Write a function to find the correct number of answers.
   b. Write a function to sort the answer keys.
   c. Write a function to find the statistical frequency of each letters.
   d. Create a menu for the above operations to perform any of the functions whenever needed and the menu will exit if 'x' or 'X' is pressed.

University of Dhaka

Institute of Information Technology (IIT)

BSSE1st Year 1stSemester Midterm Examination-2024

Course #GE 1105 (Introduction to Sociology)

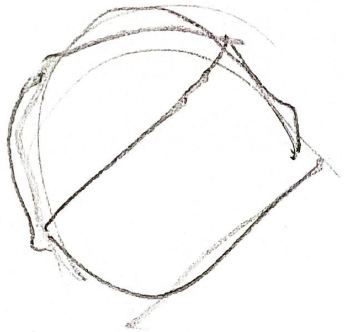
Marks-20 Time-01 Hour

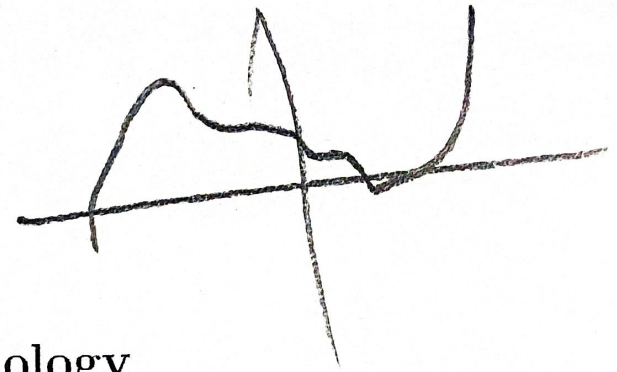
[Answer any two of the following questions in English. All questions carry equal marks]

1. *"Society is a web of social relationships"*-Explain the quote given by MacIver in your own words. Discuss the main features of the different stages of social evolution.

2. How is enlightenment related to the development of Sociology? Discuss the importance of studying sociology in the present social context.

3. What do you mean by sociological perspective? Explain the major sociological perspectives with suitable examples.





University of Dhaka

Institute of Information Technology

First Year First Semester (2024) Incourse Examination

MATH 1107

Time: 1 Hour Full Marks: 20

N.B. Answer all the questions with appropriate reasoning.

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1. For the function $h(x) = 3x^5 - 5x^3 + 3$ identify the intervals where the function is increasing and decreasing and the intervals where the function is concave up and concave down. Use this information to sketch the graph. [6]

2. In the following assume that x, y and z are all functions of t. Given $x = 4$, $y = -2$, $z = 1$, and $\frac{dx}{dt} = 9$, $\frac{dy}{dt} = -3$. Determine $\frac{dz}{dt}$ for the following equation $x(1-y)+5z^3 = y^2z^2+x^3-3$. [3]

3. A tank of water in the shape of a cone is being filled with water at a rate of $12\frac{m^3}{sec}$. The base radius of the tank is 26 meters and the height of the tank is 8 meters. At what rate is the depth of the water in the tank changing when the radius of the top of the water is 10 meters? [6]

4. A window is being built and the bottom is a rectangle and the top is a semicircle. If there is 12 meters of framing materials what must the dimensions of the window be to let in the most light? [5]

Midterm Examination for CSE 102: Discrete Mathematics,
Bachelor of Science in Software Engineering
Institute of Information Technology, University of Dhaka
Total Marks: 20, Duration: 1 Hour

Answer all the questions.

1 Define a proposition. What are the contrapositive, the converse, and the inverse of the conditional statement "If Muhammad learns discrete mathematics, then he will find a good job" 4

2 Show that $\neg (p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences. 4

3 Show that the premises "A student in this class has not read the Discrete ebook," and "Everyone in this class passed the Midterm exam " imply the conclusion "Someone who passed the Midterm exam has not read the Discrete ebook" 4

4 Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction. 4

5 Find all minterms and the CNF expression from the following truth table. 4

P	Q	R	*f*
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	F
F	T	T	F
F	T	F	F
F	F	T	T
F	F	F	F

